

MSc

Course title: Bayes Methods
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1. Let $x_1, x_2, x_3, \dots$ be an infinite sequence of binary random variables.

(a) [2 marks] What does it mean to say that these variables are exchangeable?

Solution: (bookwork) The sequence is exchangeable if the distributions $p(x_1, x_2, \dots, x_n)$ satisfy

$$p(x_1, x_2, \dots, x_n) = p(x_{\sigma_1}, x_{\sigma_2}, \dots, x_{\sigma_n})$$

for each $n \geq 1$ and any permutation $\sigma = (\sigma_1, \sigma_2, \dots, \sigma_n)$ of the numbers $(1, 2, \dots, n)$.

(b) [3 marks] What form does de Finetti's theorem give for the joint distribution $p(x_1, x_2, \dots, x_n)$?

Solution: (bookwork) There exists a distribution function $F(\theta)$ on $[0, 1]$ such that

$$p(x_1, \dots, x_n) = \int_0^1 \left[\prod_{i=1}^n p(x_i | \theta) \right] dF(\theta)$$

where $p(x_i | \theta) = \theta^{x_i} (1 - \theta)^{1-x_i}$ and if

$$\Theta = \lim_{N \rightarrow \infty} N^{-1} \sum_{i=1}^N X_i \quad \text{then} \quad F(\theta) = \Pr(\Theta \leq \theta).$$

(c) [3 marks] Write down the first three of the Savage axioms, in the form given by de Groot.

Solution: (bookwork) Let S be a sample space and let $\mathcal{S}$ be a σ -field of sets in S . In the following events A, B etc are subsets of S and members of $\mathcal{S}$.

Axiom 1. For any two events A and B exactly one of the following relations must hold: $A > B$, $A < B$, $A \sim B$.

Axiom 2. If $A_1 \cap A_2 = B_1 \cap B_2 = \emptyset$ and $A_i \geq B_i, i = 1, 2$ then $A_1 \cup A_2 \geq B_1 \cup B_2$. If in addition either $A_1 > B_1$ or $A_2 > B_2$ then $A_1 \cup A_2 > B_1 \cup B_2$.

Axiom 3. If $A \in \mathcal{S}$ then $\emptyset \leq A$.

(d) [6 marks] John elicits a separate prior $p_n(x_1, x_2, \dots, x_n)$ for each value of $n = 1, 2, \dots$. John's priors are not marginally consistent, that is

$$p_n(x_1, \dots, x_n) \neq p_{n+1}(x_1, \dots, x_n, 0) + p_{n+1}(x_1, \dots, x_n, 1).$$

(i) Show that, under John's priors, the variables $x_1, x_2, x_3, \dots$ do not form an infinite exchangeable sequence.

Solution: (variation of PS problem) In short, infinite exchangeable sequences are marginally consistent and this sequence is not MC, so it cannot be exchangeable. Proof: By de Finetti, the variables form an exchangeable sequence iff

$$p_n(x_1, \dots, x_n) = \int \prod_{i=1}^n p(x_i | q) dG(q)$$

for each $n = 1, 2, \dots$, so

$$p_{n+1}(x_1, \dots, x_n, x_{n+1}) = \int \prod_{i=1}^{n+1} p(x_i|q) dG(q).$$

But then

$$\begin{aligned} p_{n+1}(x_1, \dots, x_n, 0) + p_{n+1}(x_1, \dots, x_n, 1) &= \int [p(1|q) + p(0|q)] \prod_{i=1}^n p(x_i|q) dG(q) \\ &= \int \prod_{i=1}^n p(x_i|q) dG(q) \\ &= p_n(x_1, \dots, x_n) \end{aligned}$$

is a necessary condition for exchangeability, which John's priors do not satisfy. [Thought - if the sequence wasn't infinite John might be OK - could ask for an example of a finite ES that isn't consistent - is that possible?]

(ii) Show that John's priors do not satisfy the Savage axioms.

Solution: (new) Let $A = \{(x_1, \dots, x_{n+1}) \in \{0, 1\}^{n+1}\}$ and for $z = 0, 1$, let

$$A_z = \{(x_1, \dots, x_{n+1}) \in \{0, 1\}^{n+1}; x_{n+1} = z\}.$$

We must have either

$$p_n(x_1, \dots, x_n) < p_{n+1}(x_1, \dots, x_n, 0) + p_{n+1}(x_1, \dots, x_n, 1)$$

or

$$p_n(x_1, \dots, x_n) > p_{n+1}(x_1, \dots, x_n, 0) + p_{n+1}(x_1, \dots, x_n, 1).$$

Suppose $p_n(x_1, \dots, x_n) > p_{n+1}(x_1, \dots, x_n, 0) + p_{n+1}(x_1, \dots, x_n, 1)$. This corresponds to the preference $A > A_0 \cup A_1$. But (as sets) $A = A_0 \cup A_1$ so $A \sim A_0 \cup A_1$. This contradicts the first axiom since exactly one of $>, <, \sim$ must hold. The other possibility leads to $A < A_0 \cup A_1$ and $A \sim A_0 \cup A_1$.

(e) [6 marks] Jane elicits a process generating $x_1, x_2, x_3, \dots$ in which $x_1 = 1$ with probability p and for $n = 1, 2, \dots$,

$$p(x_{n+1} = 1 | x_1, \dots, x_n) = \frac{p + k_n}{1 + n},$$

where $k_n = \sum_{i=1}^n x_i$. Show that this process generates an infinite exchangeable sequence.

Solution: (new) If $q \sim \text{Beta}(\alpha, \beta)$ and $x_i = 1$ wp q and otherwise $x_i = 0$ iid for $i = 1, \dots, n$ then

$$p(x_1, \dots, x_n | q) = p^k (1 - p)^{n-k}$$

where $k = \sum_i x_i$. It follows that

$$\begin{aligned} p(x_1, \dots, x_n) &= \int q^{k_n} (1-q)^{n-k_n} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} q^{\alpha-1} (1-q)^{\beta-1} dq \\ &= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \frac{\Gamma(\alpha+k)\Gamma(\alpha+\beta+n-k)}{\Gamma(\alpha+\beta+n)} \end{aligned}$$

and so

$$\frac{p(x_1, \dots, x_n, 1)}{p(x_1, \dots, x_n)} = \frac{\alpha+k}{\alpha+\beta+n}.$$

Now $\Pr(x_1 = 1) = p$ so $p = E(E(x_1|q)) = \alpha/(\alpha+\beta)$ so if we take $\alpha = p$ and $\beta = 1-p$ then we match this and

$$p(x_n = 1 | x_1, \dots, x_n) = \frac{(p+k_n)}{1+n}.$$

It follows by de Finetti that $x_1, x_2, \dots$ is an infinite exchangeable sequence since

$$p(x_1, \dots, x_n) = \int \prod_{i=1}^n q^{x_i} (1-q)^{1-x_i} \text{Beta}(q; p, 1-p) dq.$$

2. Bees visit a flower at times $T_0, T_1, \dots, T_n$ with n fixed. Let $y_i = T_i - T_{i-1}, i = 1, 2, \dots, n$. Are these intervals exponentially distributed? In model $m = 1$, $y_i \sim \text{Exp}(\beta)$, with unknown rate $\beta > 0$. Bees may avoid a flower already occupied. In model $m = 2$ the intervals $y_i \sim \text{Gamma}(\alpha, \beta)$ have a Gamma distribution with unknown $\alpha > 1$ and $\beta > 0$. Experience in other studies shows μ is a central value for the interval distribution, and intervals shorter than τ and longer than τ' are improbable.

- (a) [4 marks] Briefly outline the factors to consider in eliciting priors for the model index m and the model parameters α and β . Why is $\alpha > 1$ in the second model?

Solution: (standard application of BW) The hypothesis concerns the choice of model. It is natural to take a prior $\pi_M(m) = 1/2$ which is non-informative with respect to the hypothesis. For the β -prior, under model 1, we have knowledge of a function of the parameter $1/\beta$. We can choose the prior hyper-parameters so that $E(1/\beta) = \mu$, and at the same time set the 5% quantile of the prior predictive distribution for y to some value equal to or close to τ . If the prior predictive distribution is not tractable, simulation may be used to check and adjust prior parameters. We focus on the lower tail since the hypothesis predicts a difference in the lower tail. Considering the second model, since we are testing for $\alpha = 1$, a prior for α that puts a large weight on large values of α is unrealistic and likely to lead to rejecting the second model. Care must be taken to restrict α and β jointly to physically plausible values. If we take an exponential prior for α conditioned to be greater than 1, and a Gamma prior for β , then the lower 5% and upper 95% quantile of the prior predictive distribution for the interval between arrivals should be set to τ and τ' respectively, and again $E(\alpha/\beta) = \mu$.

- (b) [3 marks] Let $\theta = \beta$ if $m = 1$ and let $\theta = (\alpha, \beta)$ if $m = 2$. Suppose m and θ have the joint prior $\pi(\theta, m) = \pi_M(m)\pi(\theta|m)$. Write down the joint posterior distribution $\pi(\theta, m|y)$ giving as much detail as you can, but without further prior elicitation.

Solution: (similar to PS problems)

$$\pi(\theta, m|y) = \begin{cases} [\prod_i \text{Exp}(y_i; \beta)] \pi(\beta|m=1) \pi_M(m=1)/p(y) & \text{if } m=1 \\ [\prod_i \text{Gamma}(y_i; \alpha, \beta)] \pi(\alpha, \beta|m=2) \pi_M(m=1)/p(y) & \text{if } m=2. \end{cases}$$

where

$$p(y) = \pi_M(m=1)p(y|m=1) + \pi_M(m=2)p(y|m=2),$$

with

$$p(y|m=1) = \int_0^\infty \left[\prod_i \text{Exp}(y_i; \beta) \right] \pi(\beta|m=1) d\beta$$

and

$$p(y|m=2) = \int_0^\infty \int_1^\infty \left[\prod_i \text{Gamma}(y_i; \alpha, \beta) \right] \pi(\alpha, \beta|m=2) d\alpha d\beta.$$

- (c) [3 marks] How would you use Monte Carlo samples $\theta^{(t)}, m^{(t)} \sim \pi(\theta, m|y)$, $t = 1, 2, \dots, T$ from the joint distribution to test model 1 against model 2?

Solution: (standard application like PS question) The Bayes factor

$$\begin{aligned} B_{1,2} &= \frac{p(y|m=1)}{p(y|m=2)} \\ &= \frac{\pi(m=1|y)}{\pi(m=2|y)} \times \frac{\pi_M(m=2)}{\pi_M(m=1)}. \end{aligned}$$

Large values of $B_{1,2} \gg 1$ are evidence for model 1 and correspondingly for $B_{2,1}$. These are reported according to a standard scale. Now if $\hat{\pi}_m = T^{-1} \sum_t \mathbb{I}_{m^{(t)}=m}$ then

$$\hat{B}_{1,2} = \frac{\hat{\pi}_1}{1 - \hat{\pi}_1} \times \frac{1 - \pi_M(m=1)}{\pi_M(m=1)},$$

since the prior probabilities are known. This (ie $\hat{B}_{1,2}$) is a consistent estimator for $B_{1,2}$.

- (d) [7 marks] Give a reversible jump MCMC algorithm targeting $\pi(\theta, m|y)$, specifying the proposals and the acceptance probabilities for the updates which increase and decrease dimension and for a fixed dimension update. You may assume the marginal priors $\pi(\alpha|m=2)$ and $\pi(\beta|m=1, 2)$ are easily sampled.

Solution: (similar to a PS example) Suppose $X_t = \theta, m$ with $\theta = \beta$ if $m = 1$ and $\theta = (\alpha, \beta)$ if $m = 2$.

(A) Suppose $m = 1$. Propose a move to $m = 2$ wp 1/2. I will simulate $\alpha' \sim \pi(\alpha|m=2)$ (prior as proposal) and set $m' = 2$ and $\theta' = (\alpha', \beta)$.

(B) Suppose $m = 2$. Propose a move to $m = 1$ wp 1/2. Set $m' = 1$ and $\theta' = \beta$.

(C) if (ie wp 1/2) we do not apply (A) or (B) we do a fixed dimension update on β (which is present in both models). Simulate $\beta' \sim \pi(\beta|m)$ and set $\theta' = \beta'$ if $m = 1$ and $\theta' = (\alpha, \beta')$ if $m = 2$.

Generate new states by applying operation (A), (B) or (C). The acceptance prob-

ability in case (A) is

$$\alpha^+(\theta'|\theta) = \min \left\{ 1, \frac{\pi(\theta', m=2|y)}{\pi(\theta, m=1|y)\pi(\alpha'|m=2)} \right\}$$

The acceptance probability in case (B) is

$$\alpha^-(\theta'|\theta) = \min \left\{ 1, \frac{\pi(\theta', m=1|y)\pi(\alpha|m=2)}{\pi(\theta, m=2|y)} \right\}$$

The acceptance probability in case (C) is

$$\alpha(\theta'|\theta) = \min \left\{ 1, \frac{\pi(\theta', m|y)\pi(\beta|m)}{\pi(\theta, m|y)\pi(\beta'|m)} \right\}$$

If the state is accepted we set $X_{t+1} = \theta'$ and otherwise $X_{t+1} = \theta$.

- (e) [4 marks] Suppose we have $\theta_1^{(t)} \sim \pi(\theta|y, m=1)$ and $\theta_2^{(t)} \sim \pi(\theta|y, m=2)$ for $t = 1, 2, \dots, T$. Give an estimator for $\pi(m=1|y)$ in terms of these samples and show how it may be used to generate samples $y^{(t)}, t = 1, 2, \dots, T$ distributed approximately according to the posterior predictive distribution, $y^{(t)} \sim p(\cdot|y)$ averaged over models. Assume the likelihood $p(y|\theta, m)$ is easily sampled.

Solution: (new) We can sample $p(y'|y)$ by sampling y', θ, m jointly according to $p(y', \theta, m|y)$ and keeping y' . Since

$$\begin{aligned} p(y', \theta, m|y) &= p(y'|\theta, m)\pi(\theta, m|y) \\ &= p(y'|\theta, m)\pi(\theta|m, y)\pi(m|y) \end{aligned}$$

we can simulate $m^{(t)} \sim \pi(m|y)$, take the existing sample $\theta_{m^{(t)}}^{(t)}$ as a sample from $\pi(\theta|m^{(t)}, y)$ and then simulate $y^{(t)} \sim p(\cdot|\theta_{m^{(t)}}^{(t)}, m^{(t)})$. We have all this except $m \sim \pi(m|y)$. Since

$$\begin{aligned} \pi(m=1|y) &= \frac{p(y|m=1)\pi_M(m=1)}{p(y|m=1)\pi_M(m=1) + p(y|m=2)\pi_M(m=2)} \\ &= \frac{\pi_M(m)}{\pi_M(m=1) + \frac{p(y|m=2)}{p(y|m=1)}\pi_M(m=2)} \end{aligned}$$

and $\pi(m=2|y) = 1 - \pi(m=1|y)$, we can simulate $m \sim \pi(m|y)$ if we can estimate $\frac{p(y|m=1)}{p(y|m=2)}$. The simplest acceptable approach is the harmonic estimator,

$$\hat{p}(y|m) = \left[T^{-1} \sum_{t=1}^T p(y|\theta_m^{(t)}, m) \right]^{-1}.$$

We form this estimate for $m=1, 2$, use it to form an estimate of $\pi(m=1|y)$ and $\pi(m=2|y)$, and proceed to simulate $y^{(t)}$ as above.